Survey Methods
5.1 – Data analysis & reporting

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Stellenbosch University Iyunivesithi Universiteit
forward together
sonke siya phambili
saam vorentoe

Outline

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- Data analysis plan
- Principles of data reporting
- What to report: a checklist

Data analysis plan

Report

What do you plan to do with you data?

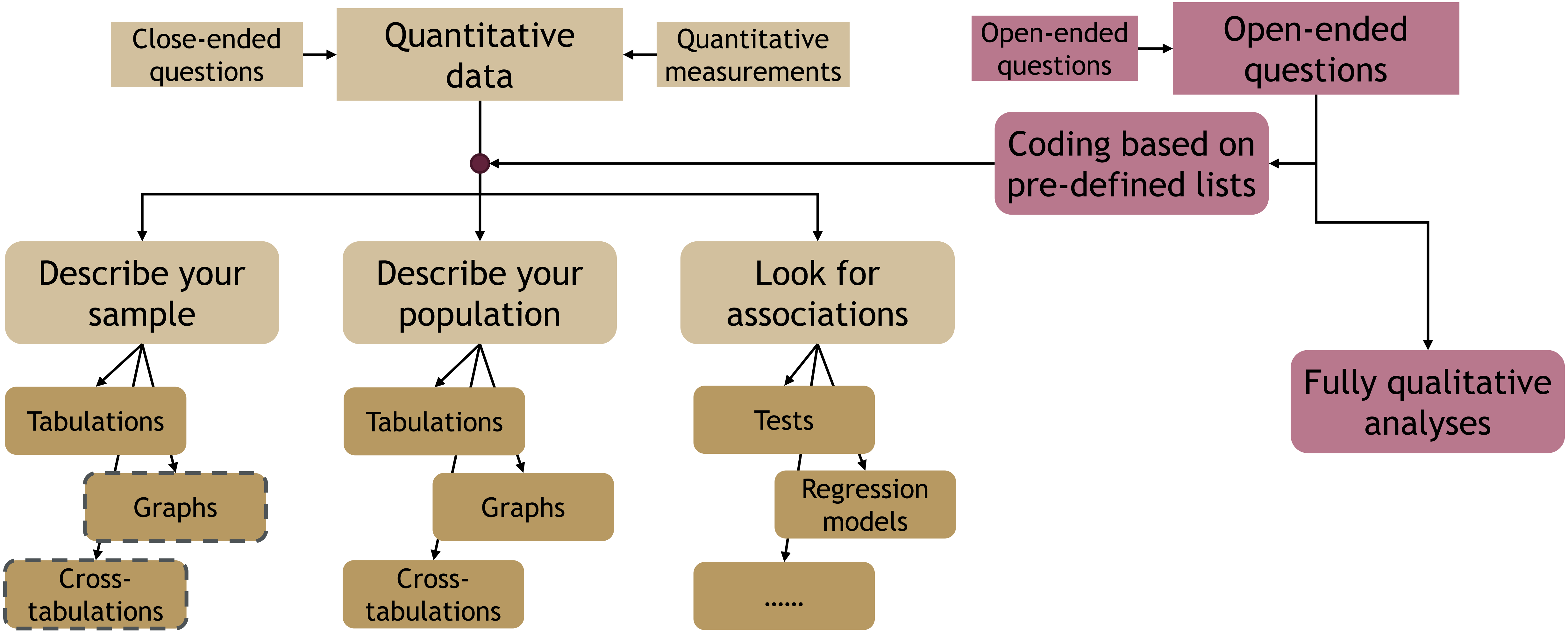
Study design

Research question

Data collection tools

Data collection methods

A (Basic) data analysis plan



Four issues in survey data analysis you should be aware of

1

ADJUSTING FOR SAMPLE NONRESPONSE AND SAMPLE FRAME DEFICIENCIES

2

DEALING WITH ITEM NONRESPONSE

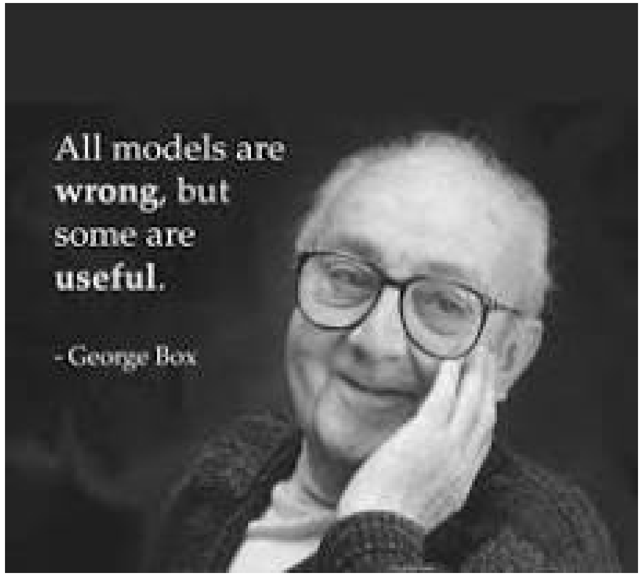
3

ADJUSTING FOR DIFFERENT PROBABILITIES OF SELECTION

4

CALCULATING SAMPLING ERRORS

Reporting



Why is important?

Understanding

Additional Materials

Analyses

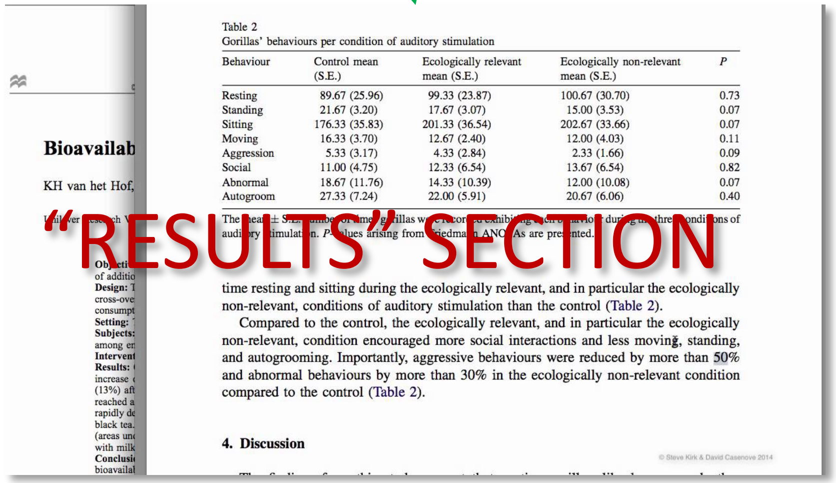


Table 2
Gorillas' behaviours per condition of auditory stimulation

Behaviour	Control mean (S.E.)	Ecologically relevant mean (S.E.)	Ecologically non-relevant mean (S.E.)	P
Resting	89.67 (25.96)	99.33 (23.87)	100.67 (30.70)	0.73
Standing	21.67 (3.20)	17.67 (3.07)	15.00 (3.53)	0.07
Sitting	176.33 (35.83)	201.33 (36.54)	202.67 (33.66)	0.07
Moving	16.33 (3.70)	12.67 (2.40)	12.00 (4.03)	0.11
Aggression	5.33 (3.17)	4.33 (2.84)	2.33 (1.66)	0.09
Social	11.00 (4.75)	12.33 (6.54)	13.67 (6.54)	0.82
Abnormal	18.67 (11.76)	14.33 (10.39)	12.00 (10.08)	0.07
Autogroom	27.33 (7.24)	22.00 (5.91)	20.67 (6.06)	0.40

time resting and sitting during the ecologically relevant, and in particular the ecologically non-relevant, conditions of auditory stimulation than the control (Table 2).

Compared to the control, the ecologically relevant, and in particular the ecologically non-relevant, condition encouraged more social interactions and less moving, standing, and autogrooming. Importantly, aggressive behaviours were reduced by more than 50% and abnormal behaviours by more than 30% in the ecologically non-relevant condition compared to the control (Table 2).

4. Discussion

- Sample characteristics

  “Table 1”

  Response rates

- Answer your research questions
  - Descriptive results (tables, ...)
  - Differences between subpopulations (tests, cross tabs, plots...)
  - Relationships between variables (correlations, regression, ...)

Table 1 Sample descriptive statistics at baseline[a]

Variable	*N*	Median/percentage	IQR/frequency	Range
Men	10100	37.81 %	3819	
Age [years]	10099	38	[26 ; 52]	[18 ; 101]
Race	10100			
Black		81.07 %	8188	
Coloured		14.31 %	1445	
White		3.38 %	341	
Asian		1.25 %	126	
Education	10091			
None		15.61 %	1575	
Primary (1 – 7 years)		25.07 %	2530	
Secondary (8 – 12 years)		51.17 %	5164	
Tertiary (>12 years)		8.15 %	822	
Urban	10100	54.02 %	5456	
Quintile of household income per capita	10100			
I (0-217 ZAR[b])		15.39 %	1554	
II (218 - 345 ZAR)		16.82 %	1669	
III (346 - 531 ZAR)		18.82 %	1901	
IV (532 - 992 ZAR)		22.84 %	2307	
V (993 - 62343 ZAR)		26.13 %	2639	
Urban	10100	54.02 %	5456	
Current smoking	9138	20.09 %	1845	
Current use of alcohol	9191	22.48 %	2066	
Exercise frequency	9159			
Low (<once a week)		79.40 %	7272	
Moderate (1/2 times a week)		9.95 %	911	
High (>2 times a week)		10.66 %	976	
Waist circumference [cm]	8376	85.55	[76.03 ; 98.05]	[30.40 ; 197.30]
BMI [kg/m^2]	8177	25.18	[21.48 ; 30.85]	[10.61 ; 59.22]
BMI category	8177			
Underweight		5.38 %	440	
Normal weight		43.44 %	3552	
Overweight		23.08 %	1887	
Obese		28.10 %	2298	

[a] *N* = number of non-missing values; IQR = interquartile range
Values are unweighted
[b] ZAR = South African Rand. 1 ZAR = 0.12 US Dollars (average exchange rate in 2008)

Cois and Day *BMC Obesity* (2015) 2:42
DOI 10.1186/s40608-015-0072-2

Obesity trends and risk factors in the South African adult population

Table 3 Parameter estimates for the regression of the slope and intercept of the growth line on subjects' baseline characteristics

	Slope [kg/m^2 per decade]		Intercept [kg/m^2]	
Independent variable	Coefficient	95 % CI	Coefficient	95 % CI
Sex				
Male vs female	*−1.91*	(−2.93 ; −0.00)	*−1.54*	(−1.88 ; −1.21)
Age				
10 year increase	*−0.90*	(−1.20 ; −0.50)	−0.03	(−0.15 ; 0.09)
Race				
Coloured vs. Black	−0.51	(−1.98 ; 0.96)	−0.44	(−0.90 ; 0.03)
White vs. Black	*2.58*	(0.46 ; 4.71)	−0.35	(−1.30 ; 0.60)
Asian vs. Black	0.05	(−1.13 ; 1.0)	*−0.85*	(−1.36 ; -0.33)
Education				
None vs. primary	−0.18	(−1.69 ; 1.34)	−0.27	(−0.75 ; 0.21)
Secondary vs. primary	−0.05	(−1.13 ; 1.04)	0.17	(−0.19 ; 0.53)
Tertiary vs. primary	0.67	(−0.82 ; 2.16)	0.17	(−0.38 ; 0.72)
Income Quintile				
I vs. III	0.00	(−1.26 ; 1.26)	−0.18	(−0.60 ; 0.25)
II vs. III	0.51	(−0.84 ; 1.86)	−0.16	(−0.58 ; 0.25)
IV vs. III	1.28	(−0.08 ; 2.64)	−0.26	(−0.63 ; 0.10)
V vs. III	*1.59*	(0.41 ; 2.78)	−0.21	(−0.58 ; 0.17)

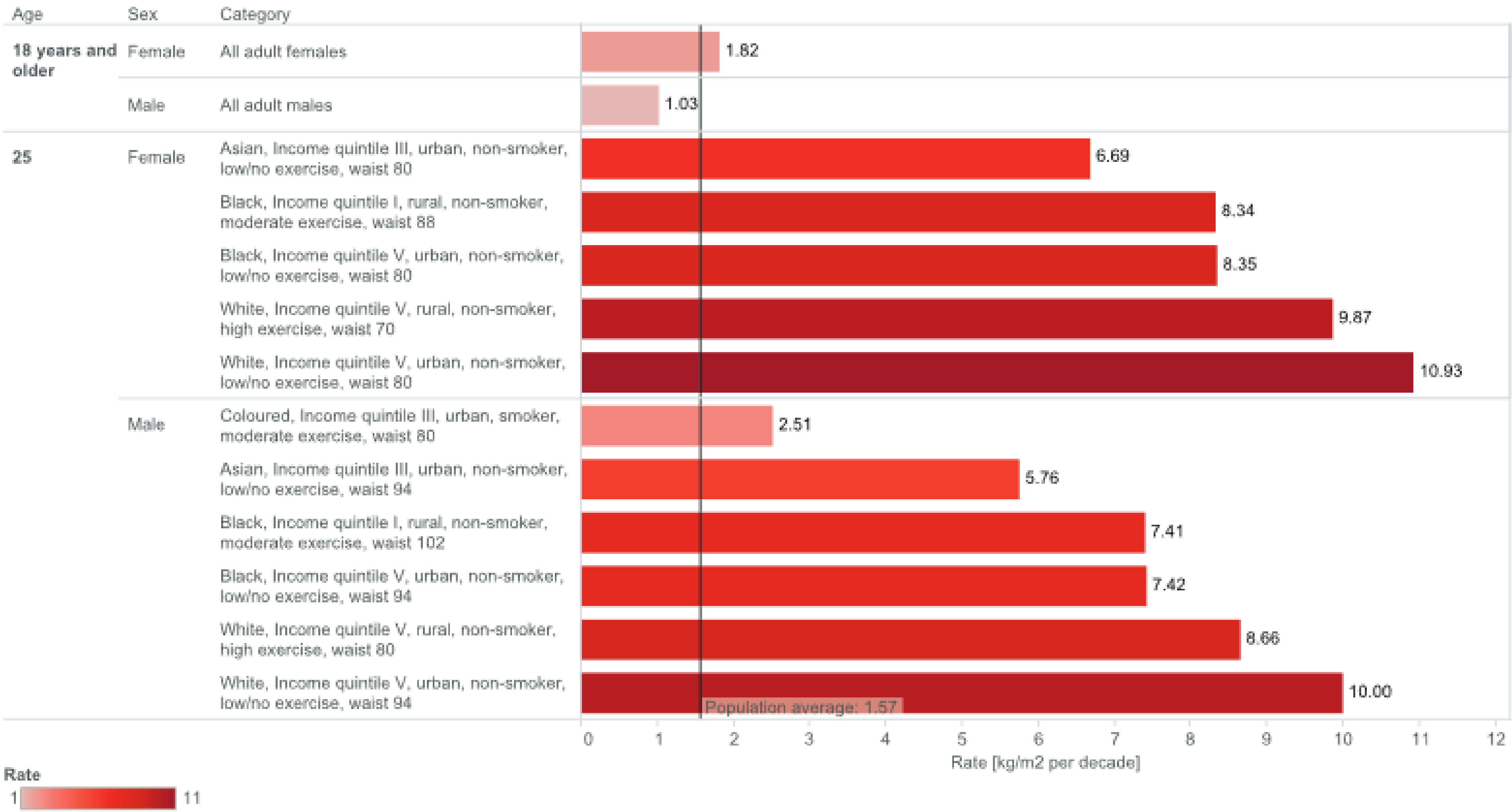


Fig. 1 Estimated rate of increase in BMI in selected high-risk sub-groups of the South African adult population between 2008 and 2012. Values represent the average rate of increase in BMI for individuals with normal weight at baseline and the combination of sociodemographic and bio-behavioural characteristics indicated in the figure, estimated form the multivariate model described in the text. The values of the variables not explicitly indicated in the figure are set to the population average

Beyond the results sections: a full outline

- Results - describing what the research revealed

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- Introduction - setting the stage for the report (background, literature review, rationale)
- Aims and objectives – what did you want to do? The research questions
- Methods - explaining how the research was conducted
- Results - describing what the research revealed
- Discussion - analysing the results in the context of what was already known to give a full answer to your questions (or reasons why you could not…)
- Limitations - revealing any research oversights or areas for further research, and speculate how these limitations could have affected your conclusions
- Conclusion - a summary of the research analysis and why your readers should care about it

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Beyond the results sections: a full outline

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  - Executive summary
- Aims and objectives – what did you want to do? The research questions
- Methods - explaining how the research was conducted
- Results - describing what the research revealed
- Discussion - analysing the results in the context of what was already known to give a full answer to your questions (or reasons why you could not…)
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In summary: The content of a survey report

Background, rationale, research questions

Why you undertook your study

Methods

What you did

What you could not do

Limitations

Results

What you found

Discussion, conclusions

Why this is important

A checklist

JGIM

RESEARCH METHODS

A Consensus-Based Checklist for Reporting of Survey Studies (CROSS)

(Sharma et al. 2021)

Supplement 3: Checklist for Reporting Of Survey Studies (CROSS)

Section/topic	Item	Item description	Reported on page #
Title and abstract			
Title and abstract	1a	State the word “survey” along with a commonly used term in title or abstract to introduce the study’s design.	
	1b	Provide an informative summary in the abstract, covering background, objectives, methods, findings/results, interpretation/discussion, and conclusions.	
Introduction			
Background	2	Provide a background about the rationale of study, what has been previously done, and why this survey is needed.	
Purpose/aim	3	Identify specific purposes, aims, goals, or objectives of the study.	
Methods			
Study design	4	Specify the study design in the methods section with a commonly used term (e.g., cross-sectional or longitudinal).	
	5a	Describe the questionnaire (e.g., number of sections, number of questions, number and names of instruments used).	

	5b	Describe all questionnaire instruments that were used in the survey to measure particular concepts. Report target population, reported validity and reliability information, scoring/classification procedure, and reference links (if any).
Data collection methods	5c	Provide information on pretesting of the questionnaire, if performed (in the article or in an online supplement). Report the method of pretesting, number of times questionnaire was pre-tested, number and demographics of participants used for pretesting, and the level of similarity of demographics between pre-testing participants and sample population.
	5d	Questionnaire if possible, should be fully provided (in the article, or as appendices or as an online supplement).
	6a	Describe the study population (i.e., background, locations, eligibility criteria for participant inclusion in survey, exclusion criteria).
Sample characteristics	6b	Describe the sampling techniques used (e.g., single stage or multistage sampling, simple random sampling, stratified sampling, cluster sampling, convenience sampling). Specify the locations of sample participants whenever clustered sampling was applied.
	6c	Provide information on sample size, along with details of sample size calculation.
	6d	Describe how representative the sample is of the study population (or target population if possible), particularly for population-based surveys.

Survey administration	7a	Provide information on modes of questionnaire administration, including the type and number of contacts, the location where the survey was conducted (e.g., outpatient room or by use of online tools, such as SurveyMonkey).
	7b	Provide information of survey's time frame, such as periods of recruitment, exposure, and follow-up days.
	7c	Provide information on the entry process: –>For non-web-based surveys, provide approaches to minimize human error in data entry. –>For web-based surveys, provide approaches to prevent "multiple participation" of participants.
Study preparation	8	Describe any preparation process before conducting the survey (e.g., interviewers' training process, advertising the survey).
Ethical considerations	9a	Provide information on ethical approval for the survey if obtained, including informed consent, institutional review board [IRB] approval, Helsinki declaration, and good clinical practice [GCP] declaration (as appropriate).
	9b	Provide information about survey anonymity and confidentiality and describe what mechanisms were used to protect unauthorized access.

Statistical analysis	10b	Report any modification of variables used in the analysis, along with reference (if available).
	10c	Report details about how missing data was handled. Include rate of missing items, missing data mechanism (i.e., missing completely at random [MCAR], missing at random [MAR] or missing not at random [MNAR]) and methods used to deal with missing data (e.g., multiple imputation).
	10d	State how non-response error was addressed.
	10e	For longitudinal surveys, state how loss to follow-up was addressed.
	10f	Indicate whether any methods such as weighting of items or propensity scores have been used to adjust for non-representativeness of the sample.
	10g	Describe any sensitivity analysis conducted.

Results		
Respondent characteristics	11a	Report numbers of individuals at each stage of the study. Consider using a flow diagram, if possible.
	11b	Provide reasons for non-participation at each stage, if possible.
	11c	Report response rate, present the definition of response rate or the formula used to calculate response rate.
	11d	Provide information to define how unique visitors are determined. Report number of unique visitors along with relevant proportions (e.g., view proportion, participation proportion, completion proportion).
Descriptive results	12	Provide characteristics of study participants, as well as information on potential confounders and assessed outcomes.
Main findings	13a	Give unadjusted estimates and, if applicable, confounder-adjusted estimates along with 95% confidence intervals and p-values.
	13b	For multivariable analysis, provide information on the model building process, model fit statistics, and model assumptions (as appropriate).
	13c	Provide details about any sensitivity analysis performed. If there are considerable amount of missing data, report sensitivity analyses comparing the results of complete cases with that of the imputed dataset (if possible).

Discussion		
Limitations	14	Discuss the limitations of the study, considering sources of potential biases and imprecisions, such as non-representativeness of sample, study design, important uncontrolled confounders.
Interpretations	15	Give a cautious overall interpretation of results, based on potential biases and imprecisions and suggest areas for future research.
Generalizability	16	Discuss the external validity of the results.
Other sections		
Role of funding source	17	State whether any funding organization has had any roles in the survey's design, implementation, and analysis.
Conflict of interest	18	Declare any potential conflict of interest.
Acknowledgements	19	Provide names of organizations/persons that are acknowledged along with their contribution to the research.

Summary questions

- What is a data analysis plan?
- Basic concepts of quantitative data analysis
- Issues in analysing survey data
- How to report survey studies

References

Babbie, Earl R. 2020. *The Practice of Social Research*. Cengage learning. Chapter 14.

Fowler Jr, Floyd J. 2013. *Survey Research Methods*. Sage publications. Chapter 10.

Sharma, Akash, Nguyen Tran Minh Duc, Tai Luu Lam Thang, Nguyen Hai Nam, Sze Jia Ng, Kirellos Said Abbas, Nguyen Tien Huy, et al. 2021. 'A Consensus-Based Checklist for Reporting of Survey Studies (CROSS)'. *Journal of General Internal Medicine* 36 (10): 3179–87. https://doi.org/10.1007/s11606-021-06737-1.

Other resources

Claire Irwin, and Erin Stafford. 2022. 'Survey Methods for Educators'. Indexes; Offices. Regional Educational Laboratory Program (REL). 2022. https://ies.ed.gov/ncee/rel/Products/Publication/3752.

Thank you

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